

SMCS - IBA CSE248 - CCN Final Exam Spring 2024

Institute of Business Administration

Course: Comp. Communications & Networking	Exam: Final
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Class: BSCS	Date: 18 th May, 2024

Name:

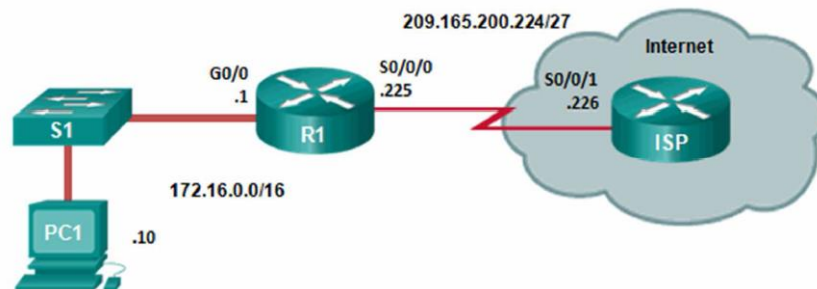
ERP No:

PART B - SUBJECTIVE (80 Points) (Time 150 Minutes)

Short Question/Answer Section: (Attempt all, each has 2 points) 20 points

1. In an IP Header, a packet has arrived in which the offset value is 100, the value of HLEN is 5, and the value of the total length field is 100. What are the numbers of the first byte and the last byte?

2. R1 was configured with the static route command `ip route 209.165.200.224 255.255.255.224 S0/0/0` and consequently users on network 172.16.0.0/16 are unable to reach resources on the Internet. How should this static route be changed to allow user traffic from the LAN to reach the Internet?



3. Assume a TCP client expects to receive byte 2001, but it receives a segment with sequence number 1201. What is the reaction of the TCP client to this event? Can you justify the reaction?

4. Suppose that N switches supporting K VLAN groups are to be connected via a trunking protocol. How many ports are needed to connect the switches? Justify your answer

5. Discuss why each input port in a high-speed router stores a shadow copy of the forwarding table.

6. What is HOL blocking? Does it occur in input ports or output ports?

7. What field in the IP header can be used to ensure that a packet is forwarded through no more than N routers?

8. Suppose there are three routers between a source host and a destination host. Ignoring fragmentation, an IP datagram sent from the source host to the destination host will travel over how many interfaces? How many forwarding tables will be indexed to move the datagram from the source to the destination?

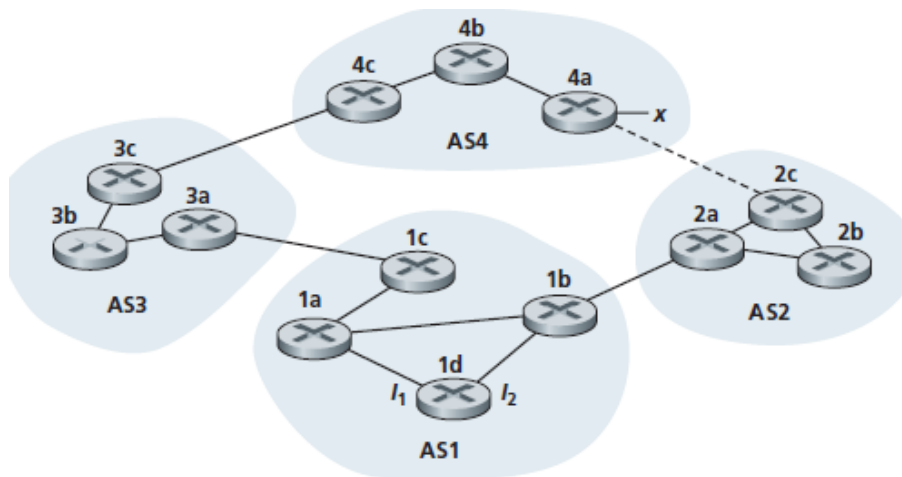
9. Suppose an application generates chunks of 40 bytes of data every 20 msec, and each chunk gets encapsulated in a TCP segment and then an IP datagram. What percentage of each datagram will be overhead, and what percentage will be application data?

10. Compare and contrast the IPv4 and the IPv6 header fields. Do they have any fields in common?

Detailed Question/Answers Section (Attempt all questions) 60 points

1. Consider the network shown below. Suppose AS3 and AS2 are running OSPF for their intra-AS routing protocol. Suppose AS1 and AS4 are running RIP for their intra-AS routing protocol. Suppose eBGP and iBGP are used for the inter-AS routing protocol. Initially suppose there is *no* physical link between AS2 and AS4. (4)

- Router 3c learns about prefix x from which routing protocol? →
- Router 3a learns about x from which routing protocol? →
- Router 1c learns about x from which routing protocol? →
- Router 1d learns about x from which routing protocol? →



2. Show the functioning of IEEE802.11-MAC (WiFi) through a timing diagram showing its features like RTS/CTS/ACK/ NAV using sender node, receiving node and hidden nodes, with respect to sender node, by denoting them with S, R and H1 and H2 respectively. (4)

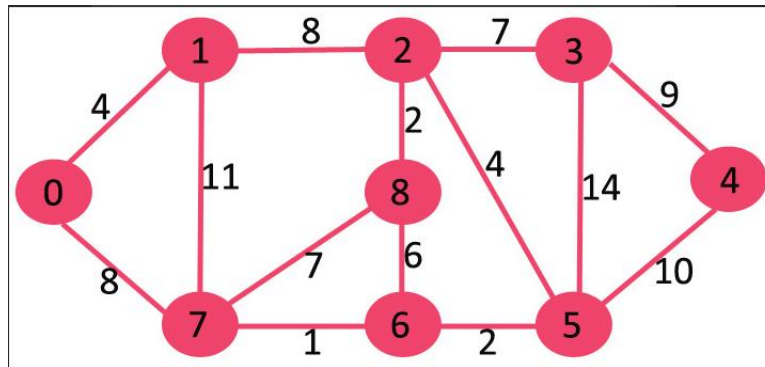
H1

R

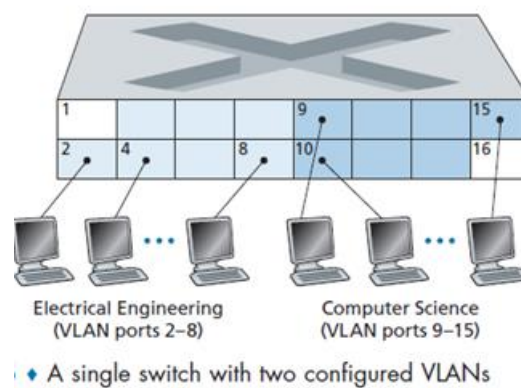
S

H2

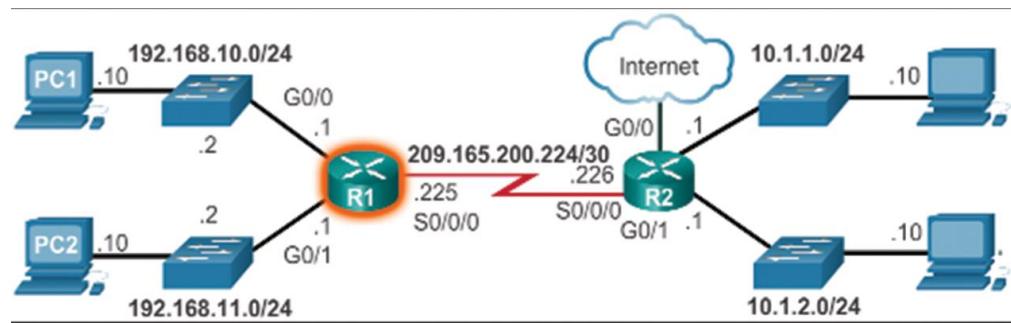
3. Consider the following network. With the indicated link costs, use Dijkstra's shortest-path algorithm to compute the shortest path from 7 to all network nodes. Show how the algorithm works by computing a complete table. (5)



4. Consider the single switch VLAN in Figure below and assume an external router is connected to switch port 1. Assign IP addresses to the EE and CS hosts and router interface. Trace the steps taken at both the network layer and the link layer to transfer an IP datagram from an EE host to a CS host (4)



5. Make a routing table for Router R1 & R2 using the configuration given in the figure. (6)



Complete Routing table for router R1

Mask	Network Address	Next Hop	Interface

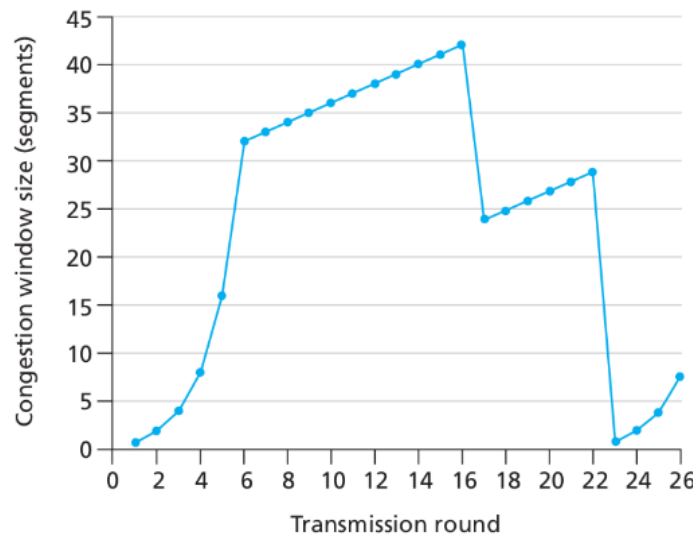
Complete Routing table for router R2

Mask	Network Address	Next Hop	Interface

6. a. Compute the checksum for the following data word: **0x 1578ab4f0c98a** (2)
- b. Also state where it is implemented practically? (2)

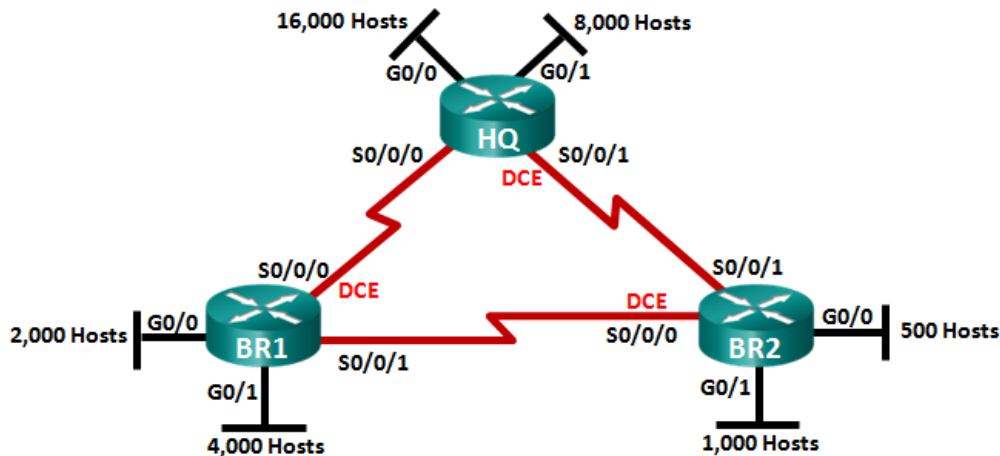
7. Consider the 5-bit generator, $G=10011$, and suppose that D has the value 1010101010 . What is the value of R ? (4)

8. Assuming TCP Reno is the protocol experiencing the behavior shown above, answer the following questions. In all cases, you should provide a short discussion justifying your answer. (12)



- Identify the intervals of time when TCP slow start is operating.
- Identify the intervals of time when TCP congestion avoidance is operating.
- After the 16th transmission round, is segment loss detected by a triple duplicate ACK or by a timeout?
- After the 22nd transmission round, is segment loss detected by a triple duplicate ACK or by a timeout?
- What is the initial value of ssthresh at the first transmission round?
- What is the value of ssthresh at the 18th transmission round?
- What is the value of ssthresh at the 24th transmission round?
- Assuming a packet loss is detected after the 26th round by the receipt of a triple duplicate ACK, what will be the values of the congestion window size and of ssthresh?
- Suppose TCP Tahoe is used (instead of TCP Reno), and assume that triple duplicate ACKs are received at the 16th round. What are the ssthresh and the congestion window size at the 19th round?

9. Given the following topology examine the network requirements and develop the VLSM address scheme using 172.16.128.0/17 network address. Populate the given tables (re-write them as it is in your answer booklet). Bear that we would adhere to the principle that first host address of the network will be assigned to the router. Likewise, HQ interfaces would have priority over others and BR1 interface over BR2 for assigning IP addresses to interfaces over serial links. Show your complete working or else your answer will be annulled (12)



Subnet Tables:

Subnet Description	Number of Hosts Needed	Network Address /CIDR	First Host Address	Broadcast Address
HQ G0/0	16,000			
HQ G0/1	8,000			
BR1 G0/1	4,000			
BR1 G0/0	2,000			
BR2 G0/1	1,000			
BR2 G0/0	500			
HQ S0/0/0 – BR1 S0/0/0	2			
HQ S0/0/1 – BR2 S0/0/1	2			
BR1 S0/0/1 – BR2 S0/0/0	2			

Device	Interface	IP Address	Subnet Mask	Device Interface
HQ	G0/0			16,000 Host LAN
	G0/1			8,000 Host LAN
	S0/0/0			BR1 S0/0/0
	S0/0/1			BR2 S0/0/1
BR1	G0/0			2,000 Host LAN
	G0/1			4,000 Host LAN
	S0/0/0			HQ S0/0/0
	S0/0/1			BR2 S0/0/0
BR2	G0/0			500 Host LAN
	G0/1			1,000 Host LAN
	S0/0/0			BR1 S0/0/1
	S0/0/1			HQ S0/0/1

10.

- a. Consider the socket pair: (41.199.222.3:80, 177.41.72.6:3022) and indicate each part succinctly and specifically (i.e., not in general terms) explaining server/host ip, service, port type etc. (2)

- b. Specify TCP and UDP headers highlighting the fields that are not required in UDP with explicit reasons. (3)
